TT-P-87D
May 29, 1978
SUPERSEDING
Fed. Spec. TT-P-87C
October 9, 1970

FEDERAL SPECIFICATION

PAINT: TRAFFIC, PREMIXED, REFLECTORIZED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers paint for retroreflective traffic markings. The paint contains low-index glass spheres, and is designed to bind an optional, additional quantity of coarser, low-index spheres which are packaged separately and dropped onto the wet paint immediately after application.

1.2 Classification.

1.2.1 Color. The premixed paint shall be of the following colors (see 6.2):

White
Yellow

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

TT-B-1325 - Beads (Glass Spheres); Retro-reflective.

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.
Fed. Test Method Std. No. 370 - Instrumental Photometric Measurements of Retroreflective Materials and Retroreflective Devices.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

49 CFR 178 - Department of Transportation (DOT) Shipping Container Specifications.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 185 - Coarse Particles in Pigments, Pastes, and Paints.
D 562 - Consistency of Paints Using the Stormer Viscosimeter.
D 711 - No-Pick-Up Time of Traffic Paint.
D 968 - Abrasion Resistance of Coatings of Paint, Varnish, Lacquer, and Related Products by the Falling Sand Method.
D 969 - Resistance of Traffic Paint to Bleeding.
D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems.
D 1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes.
D 2244 - Color Differences of Opaque Materials.
D 3272 - Vacuum Distillation of Solvents from Solvent Base Paints for Analysis.
E 97 - 45-deg, 0-deg Directional Reflectance of Opaque Specimens by Filter Photometry.
G 23 - Light- and Water-Exposure Apparatus (Carbon-Arc Type) for Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The paint (premixed with glass spheres) shall consist of pigment or pigments and vehicle so combined as to produce a paint meeting all the requirements specified herein. The glass spheres shall be low-index of refraction conforming to TT-B-1325, type I, gradation B, in the proportion of 5 pounds of glass spheres per gallon of paint.

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3.1.1 The solvent, when tested as specified in 4.4.2, shall conform to the following requirements by volume:

(a) The total of solvents with olefinic or cyclo-olefinic unsaturation shall not exceed 5 percent.
(b) The total of aromatic compounds with 8 or more carbon atoms in the molecule, except ethylbenzene, methyl benzoate, and phenyl acetate, shall not exceed 8 percent.
(c) The total of ethylbenzene, toluene, and branched-chain ketones shall not exceed 20 percent.
(d) A solvent which may be classified into more than one of the above groups shall be considered a member of the group having the lowest allowable concentration.
(e) The total of (a), (b), and (c) shall not exceed 20 percent.

3.2 Qualitative requirements.

3.2.1 Color. When tested as specified in 4.4.3, the color difference between a film of the yellow pigmented binder (without glass spheres) and color chip No. 33538 of Fed. Std. No. 595 shall be no greater than ΔE ($CIE_{L^*a^*b^*}$) = 6, allowing a ΔL^* of not more than + 2.0 units or -0.5 unit, and a Δa^* of not more than + 1.0 unit or -0.5 unit from color chip No. 33538.

3.2.2 Color stability. When tested as specified in 4.4.4, the color difference between the film of yellow pigmented binder before weathering and after weathering shall be no greater than ΔE ($CIE_{L^*a^*b^*}$) = 6, and Y value for the weathered film shall not be less than 50. The directional reflectance for the film of white pigmented binder after weathering shall be a minimum of 80.

3.2.3 Condition in container. When tested as specified in table III, the premixed paint, as received, shall not curdle, skin, liver, gel, or show hard, dry caking or settling that cannot be redispersed with a paddle to a uniform, homogeneous state.

3.2.4 Adhesion. When tested as specified in 4.4.5, a dried film of the premixed paint shall not flake, crumble, or separate from the substrate.

3.2.5 Flexibility. When tested as specified in 4.4.6, the dried film of the premixed paint shall not flake or crack to the bare metal.

3.2.6 Abrasion resistance. When tested as specified in 4.4.7, the dried film of the premixed paint shall require not less than 105 liters of sand to remove an area of the film 4 mm in diameter.

3.2.7 Water resistance. When tested as specified in 4.4.8, the dried film of the premixed paint shall show no blistering, loss of adhesion, wrinkling, or discoloration.

3.2.8 Accelerated weathering. When tested as specified in 4.4.9, the dried film of premixed paint shall show no blistering, flaking, or wrinkling. There shall not be more than a 10 percent loss in weight of the dried film after weathering.

3.2.9 Luminance factor. When tested as specified in 4.4.10, the luminance factor shall be as specified in table I.

TABLE I.

Paint	Entrance angle (degrees)	Luminance factor, minimum	
		Before weathering	After weathering
White	75	5.0	4.2
White	88	20.0	17.0
Yellow	75	4.0	3.4
Yellow	88	12.0	10.0

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3.2.10 Storage stability (partially full container). The premixed paint shall show no skinning, when tested as specified in 4.4.11.

3.3 Quantitative requirements.

3.3.1 Quantitative requirements of the pigmented binder shall be as specified in table II.

TABLE II. Quantitative requirements

Characteristics	Requirements	
	Minimum	Maximum
Pigment content, percent by weight of paint		
White	60	--
Yellow	62	--
Nonvolatile vehicle, percent by weight of vehicle	35	--
Consistency, Krebs units	65	75
Dry opacity		
White	0.88	--
Yellow	0.94	--
Directional reflectance, white only	80	--
Bleeding ratio	No. 4 (Photographic std.)	--
Water content, percent by weight of paint	--	0.5
Drying time, minutes	--	40
Fineness of grind	3	--

3.4 Quantities. The paint shall be furnished in 1-gallon, 5-gallon, 30-gallon, and 55-gallon quantities.

3.5 Special marking.

3.5.1 Direction for use and preparation of substrate in accordance with manufacturer's direction shall be shown on the reverse side of the container.

3.5.2 Application (drop-on) of the separate glass spheres shall be applied at the rate of 5 pounds of glass spheres conforming to TT-B-1325, type I, gradation A, per gallon of premixed paint. The manufacturer's instructions for the application of the glass spheres shall be shown on the reverse side of the container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery and selected at random. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

4.3 Testing of the end item.

4.3.1 Lot. The paint shall be assembled into lots as specified in MIL-STD-105. In MIL-STD-105, the words "essentially the same conditions" shall be interpreted to mean a manufacturer's batch and defined as the end product of all raw materials mixed, blended, or processed in a single operation.

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4.3.2 Sampling of the end item. For the purposes of sampling, the lot shall be expressed in units of gallons. Samples from lots shall be taken in accordance with MIL-STD-105 using inspection level S-2 and an acceptable quality level (AQL) of 2.5 percent defective. For each sample of premixed paint, the manufacturer shall supply a sample of equal volume of the pigmented binder which is representative of the sample of premixed paint taken and a 2-pound sample of the glass spheres used in the premixed paint.

4.4 Test procedures.

4.4.1 The paint shall be tested according to the test methods indicated in table III. Unless otherwise specified, all tests shall be performed at standard testing conditions which are a temperature of 23° ± 1°C and a relative humidity of 50 ± 5 percent. All test reports shall contain the individual values utilized in expressing the final result. All tests shall be evaluated for conformance to the requirements specified in section 3. Failure to pass any test, or noncompliance with any requirement, shall be cause for rejection of the sample.

TABLE III. Test methods.

Characteristic	Requirement reference	Fed. Test Method Std. No. 141	ASTM Method	Test paragraph reference
Solvent analysis	3.1.1	7356	D 3272	4.4.2
Color	3.2.1	----	D 2244	4.4.3
Color stability	3.2.2	----	D 2244, E 97	4.4.4
Condition in container	3.2.3	3011	----	---
Adhesion	3.2.4	----	----	4.4.5
Flexibility	3.2.5	----	----	4.4.6
Abrasion resistance	3.2.6	----	D 968	4.4.7
Water resistance	3.2.7	----	D 1308, Sec. 5.4	4.4.8
Accelerated weathering	3.2.8	----	G 23	4.4.9
Luminance factor	3.2.9	----	----	4.4.10
Storage stability (partially full container)	3.2.10	----	----	4.4.11
Pigment content	Table II	4021	----	---
Nonvolatile vehicle	Table II	4053	----	---
Consistency	Table II	----	D 562	---
Dry opacity	Table II	4122	----	4.4.12
Directional reflectance	Table II	----	E 97	---
Bleeding ratio	Table II	----	D 969	---
Water content	Table II	4081	----	---
Drying time	Table II	----	D 711	---
Weight of glass beads	3.1	----	D 185, Sec. 6	---
Fineness of grind	Table II	----	D 1210	---

4.4.2 Solvent analysis.

4.4.2.1 Solvent extraction. The solvent shall be extracted from the paint in accordance with ASTM D 3272.

4.4.2.2 Solvent composition. The composition of the solvent shall be determined in accordance with 4.4.2.3 and method 7356 of Fed. Test Method Std. No. 141 and evaluated for compliance with the requirement of 3.1.1.

4.4.2.3 Aromatic and oxygenated solvents. The 1.83 m (6 ft) column shall be installed and the operating conditions described in method 7356 shall be followed. About 3 microliters of the isolated distillate shall be injected and the chromatogram scanned. The aliphatic solvents will emerge within 1 minute and the complete chromatogram should develop in about 5 minutes. From the position of the peaks observed on the chromatogram, an internal standard that will be free of interference shall be selected, such as cyclopentanol or cyclohexanol. Six-tenths of a milliliter of internal standard shall be added to 3 ml of the distillate. The sample shall be analyzed according to the above procedure. Peaks emerging after 1 minute are aromatic solvents along with any oxygenated solvents that may be present. The percent of aromatic and oxygenated solvents shall be calculated as follows:

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$$\text{percent aromatic and oxygenated solvents, v/v} = \frac{A \times B}{C \times D}$$

Where: A = Percent of internal standard added (in this case, 20)
B = Area of aromatic and oxygenated solvents
C = Calibration factor for the internal standard. This factor is dependent on the internal standard used and on the performance of the chromatograph, and should be determined daily
D = Area of the internal standard (in this case cyclopentanol or cyclohexanol)

4.4.3 Color. Draw down a film of yellow pigmented binder using a wet film thickness of 0.15 mm (0.006 in) on a glass panel conforming to method 2021 of Fed. Test Method Std. No. 141. Allow to dry for 24 hours at standard conditions. Determine the color of the film of yellow pigmented binder and for color chip No. 33538 of Fed. Std. No. 595 using the procedure in ASTM D 2244 and making the calculations according to the CIE (L*a*b*) system. Evaluate for compliance with the requirement in 3.2.1.

4.4.4 Color stability. Draw down films of yellow pigmented binder and white pigmented binder at a wet film thickness of 0.15 mm (0.006 in) on separate glass panels prepared as specified in method 2021 of Fed. Test Method Std. No. 141. Allow to dry for 24 hours at standard conditions. Determine color of the film of yellow pigmented binder by using the procedure in ASTM D 2244 and making the calculations according to the CIE(L*a*b*) system. Expose both films to accelerated weathering in an enclosed carbon-arc lamp apparatus for 300 hours in accordance with ASTM G 23. Remove the panels from the weatherometer and allow them to condition for 24 hours at standard conditions. Determine the color for the weathered film of yellow pigmented binder by using the procedure in ASTM D 2244 and making the calculations according to the CIE(L*a*b*) system. Measure the directional reflectance of the weathered film of white pigmented binder using ASTM E 97. Evaluate these results for compliance with the requirements in 3.2.2.

4.4.5 Adhesion. Brush a coat of the premixed paint on a commercially available cement asbestos shingle at a spreading rate 1/ of 265 ft^2/gal and allow to dry at standard conditions for 24 hours. After 24 hours drying, cut two parallel scratches using a stylus, 25 mm (1 in) apart and penetrating to the substrate. A 25 mm (1 in) wide strip of tape with a minimum adhesion value of 25 oz/in 2/ shall be applied across the scratches, adhesive side down. Press the tape with two passes of a 2 kg (4.5 lb) rubber covered roller approximately 89 mm (3.5 in) diameter by 45 mm (1.75 in) wide. The surface of the roller shall have a Durometer hardness value of 70 to 80 range. Remove the tape in one abrupt motion and examine the film for compliance with 3.2.4.

4.4.6 Flexibility. Apply a film of the premixed paint on a tin panel in accordance with method 2012 of Fed. Test Method Std. No. 141 at a spreading rate of 265 ft^2/gal. Air-dry the panel for 24 hours at standard conditions, bake for 5 hours at a temperature of 107° + 3°C, and condition for 15 minutes at standard conditions. Place the test panel, with the coated side uppermost on a 12.5 mm (1/2 in) mandrel, at a point equidistant from the top and bottom edges of the panel, and bend the panel double in 1 second. Examine the film at the bend under a magnification of 7 diameters, using diffused light with an intensity of 45 to 55 footcandles. Cracks occurring at either end and extending no more than 6.4 mm (1/4 in) shall be disregarded. Determine compliance with 3.2.5.

1/ Spreading rate may be calculated by the following formula:

$$SR = \frac{3785\ (N)\ (S)\ (A)}{144\ (W)}$$

Where: SR = Spreading rate in ft^2/gal
N = Nonvolatile content of the coating material, expressed as a decimal fraction
S = Specific gravity, in g/ml, of the coating material
A = Area covered by the coating, in square inches
W = Weight, in g, of dry test film
Constants = 3785 ml/gal and 144 in^2/ft^2.

2/ 3M No. 353 is acceptable.

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4.4.7 Abrasion resistance. Apply a film of the premixed paint on a glass panel, measuring 165 x 432 x 6.3 mm (6-1/2 x 17 x 1/4 in) prepared as specified in method 2021 of Fed. Test Method Std. No. 141 at a spreading rate of 265 ft^2/gal. Air-dry the panel for 48 hours at standard conditions, then expose the panel to accelerated weathering for 300 hours in an enclosed carbon-arc lamp apparatus in accordance with ASTM G 23. After weathering, condition the panel for 24 hours at standard conditions. Run abrasion resistance according to ASTM D 968, except the inside diameter of the metal guide tube shall be from 19.97 to 19.95 mm, and only new, unused Ottawa sand shall be used for each panel. Evaluate for compliance with 3.2.6.

4.4.8 Water resistance. Apply a film of the premixed paint on a glass panel prepared as specified in method 2021 of Fed. Test Method Std. No. 141 at a spreading rate of 265 ft^2/gal. Air-dry the panel for 1 week at standard conditions, then immerse it for 24 hours in distilled water in accordance with ASTM D 1308, section 5.4. Remove the panel and dry at standard conditions. After two hours dry, check for blistering and wrinkling to determine compliance with 3.2.7. After 24 hours dry, check for loss of adhesion and discoloration for compliance with 3.2.7.

4.4.9 Accelerated weathering. Apply a film of the premixed paint on a glass panel, measuring 165 by 432 by 6.3 mm (6-1/2 by 17 by 1/4 in) prepared as specified in method 2021 of Fed. Test Method Std. No. 141, at a spreading rate of 107 ft^2/gal. Fifteen seconds after the drawdown, uniformly drop-on glass spheres conforming to TT-B-1325, type I, gradation A, at a ratio of 10 pounds of glass spheres per 100 square feet of surface. (The glass spheres shall be applied in accordance with recommended apparatus and procedure as described in paragraphs 6.3 and 6.4.) Air dry the panel for 48 hours at standard conditions. Weigh the panel. Expose the panel to accelerated weathering in an enclosed carbon-arc lamp apparatus in accordance with ASTM G 23 for 300 hours. Remove the panel, rinse with tap water and condition for 24 hours at standard conditions. Reweigh the panel and calculate the weight loss. Evaluate for compliance with 3.2.8.

4.4.10 Luminance factor. The luminance factor shall be determined according to Fed. Test Method Std. 370. Luminance factor shall be determined before and after weathering the test panels. The test panels shall be weathered in accordance with the method in 4.4.9.

4.4.10.1 Test panels. Use the test panels as prepared for the accelerated weathering test (see 4.4.9). The test area containing beads shall measure 140 x 381 mm (5-1/2 x 15 in).

4.4.10.2 Test conditions:

(a) Observation distance (D') - 50 ft.
(b) Entrance angle (β) - 75° and 88°.
(c) Viewing Angle (ε) - 73.67° and 86.67° respectively.
(d) Observation angle (α) - 1.33°.
(e) Presentation angle = 0°.
(f) The lateral distance (d) between the photoreceptor and the source of illumination - 13.93 inches.
(g) Receptor aperature angle (δ') = 10 minutes of arc (1/6 degree).
(h) Source aperature angle (δ) = 10 minutes of arc (1/6 degree).

4.4.10.3 Results. The value of each determination shall conform with the luminance factor requirements in table I.

4.4.11 Storage stability (partially full container). A 170 g (6 oz) sample of the paint shall be measured into a wide-mouth glass jar, 114 mm (4-1/2 in) in height and 51 mm (2 in) in diameter. The cover shall be secured on tightly and the jar inverted momentarily. The jar shall then be placed in an upright position in the dark at a temperature of 22.2° to 26.7°C (72° to 80°F). The sample shall not be agitated or disturbed until inspected. The paint shall be examined after 20 days for compliance with 3.2.10.

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4.4.12 Dry opacity. A film of the pigmented binder shall be applied to a black and white sealed chart to a dry film thickness of 0.051 mm (0.002 in). The dry opacity shall be determined according to method 4122 of Fed. Test Method Std. No. 141 for compliance with the requirements in table II.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The paint shall be furnished in 1-gallon metal cans, 5-gallon metal pails, 30-gallon steel drums, or 55-gallon steel drums, as specified (see 6.2). The metal cans, pails, and drums shall meet or exceed the requirements of DOT specifications under 49 CFR 178.

5.1.2 Commercial. The paint shall be packaged in cans, pails, or drums, as applicable, in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Four 1-gallon cans of paint, packaged as specified in 5.1, shall be packed in fiberboard boxes made from weather-resistant fiberboard with a bursting test strength of not less than 275 lbs. per square inch. The box flaps shall be secured with water-resistant adhesive applied to not less than 75 percent of the surface area of contact between the flaps, or with 3-inch wide waterproof tape applied to the full length of the seams and extending over the ends not less than 3 inches. Alternatively, wirebound, cleated plywood, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The barrier material shall be sealed at the edges with waterproof tape or adhesive. Five gallon metal pails, 30-gallon drums, or 55-gallon steel drums need no further packing.

5.2.2 Commercial. The 1-gallon paint, packaged as specified in 5.1, shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.3 Palletization. When shipments to Government depots are full car or truckload, the shipping containers shall be palletized on expendable wooden pallets, 2 or 4 way design, to facilitate handling, in accordance with normal commercial practice. The palletized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The reflectorized paint is intended for marking pavements, roadways, safety zones, etc. It is not intended for marking airfield runways.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Size of unit container (see 5.1 and 5.2).
(d) Color required (see 1.2.1).
(e) If extra glass beads are desired, 5 pounds per gallon of extra glass spheres conforming to TT-B-1325 Type I, gradation A.

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6.3 Bead dropping device. The bead dropping device 1/ (see figures 3, 3a and 3b in Interim Federal Specification TT-P-001952A) has a series of "V" shaped hoppers or gates tied together mechanically so that the gates will open when triggered. This bead retaining or holding mechanism is attached to the top of a rectangular box structure, having the inside dimensions of 140 by 381 mm and 610 mm (5-1/2 by 15 by 24 in) high. To help disperse the beads uniformly, an internal screen is provided approximately 25 mm (1 in) below the head retaining mechanism. The base of the structure is open so that the paint film plate can be inserted. Mechanical means are provided to lower the structure onto the film plate and to tilt the structure so that the base may be cleaned of paint residue from prior applications.

6.4 Bead dropping procedure. A measured amount of beads are distributed in the closed "V" shaped hoppers and visually distributed uniformly over the entire bead-retaining area. The paint film is then applied to a glass plate or other surface. The plate is inserted under the structure, and the structure is lowered onto the film plate. After the specified time, the bead-retaining device is mechanically triggered allowing beads to drop uniformly to the painted surface.

6.5 $CIE_{L^*a^*b^*}$ calculations.

$$\Delta E = [(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2]^{1/2}$$

Where $L^* = 25(100Y/Y_o)^{1/3} - 16; \ 1 \leq Y \leq 100$

$a^* = 500\ [(X/X_o)^{1/3} - (Y/Y_o)^{1/3}]$

$b^* = 200\ (Y/Y_o)^{1/3} - (Z/Z_o)^{1/3}$

X_o, Y_o, Z_o define the color of the nominally white object--color stimulus, with $Y_o = 100$.

6.6 Shelf-life surveillance. For the purpose of shelf-life surveillance, the premixed paint stored in the original, unopened container shall show no skinning, livering, hard, dry caking, or gummy sediment, and shall readily remix to a homogeneous state. In addition, it shall meet the drying time requirement in table II.

DOD coordination has been waived.

Preparing Activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

DOT-FAA
NASA-JFK

1/ Mechanical drawings of the bead dropper can be obtained from GSA-FSS, Chemicals and Paints Division, FMBP, Crystal Mall, Building 4, Washington, DC 20406.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price .70 cents each.

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